



OPEN Multimodal generative adversarial networks for piano fingering correction and performance expressiveness modeling through audio-visual feature fusion

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Piano performance analysis demands sophisticated understanding of both acoustic outputs and physical gestures, yet existing computational approaches typically treat these modalities in isolation. This study presents a novel framework that leverages generative adversarial networks to jointly model piano fingering correction and performance expressiveness through integrated audio-visual analysis. We develop a hierarchical attention-based fusion mechanism that learns adaptive correspondences between acoustic events and hand movements, combined with a dual-stream GAN architecture that concurrently handles fingering classification and expressiveness assessment. Experiments on a multimodal dataset comprising 3,847 performances demonstrate that our approach achieves 89.7% frame-level fingering accuracy and maintains correlation coefficients exceeding 0.85 for dynamic expressiveness features, substantially outperforming single-modality baselines. The system provides near real-time feedback with 180ms processing latency per second of audio, enabling practical deployment in interactive music education environments. These results validate the efficacy of cross-modal deep learning for capturing the coupled relationship between biomechanical execution and artistic interpretation in skilled musical performance.

Keywords Piano performance analysis, Multimodal fusion, Generative adversarial networks, Fingering correction, Performance expressiveness, Audio-visual learning

Abbreviations

GAN	Generative Adversarial Network
MFCC	Mel-Frequency Cepstral Coefficients
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
TCN	Temporal Convolutional Network
HMM	Hidden Markov Model
cGAN	Conditional Generative Adversarial Network
STFT	Short-Time Fourier Transform
MSE	Mean Squared Error
IOI	Inter-Onset Interval
RGB	Red-Green-Blue
GPU	Graphics Processing Unit

Piano performance represents a complex sensorimotor skill that demands years of deliberate practice and expert guidance. Traditional piano pedagogy heavily relies on one-on-one instruction, where teachers manually observe students' hand positions, listen to tonal quality, and provide corrective feedback. However, the scarcity of qualified instructors and the growing demand for accessible music education have created an urgent need for intelligent auxiliary teaching systems that can partially automate the assessment and correction process¹. Recent advances in artificial intelligence, particularly in audio signal processing and computer vision, have opened

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new possibilities for developing comprehensive piano performance analysis systems that operate beyond the constraints of traditional teaching environments.

The integration of multimodal sensing technologies has fundamentally transformed how we capture and analyze musical performances. Audio analysis techniques, ranging from onset detection to timbre characterization, have matured considerably over the past decade². Simultaneously, vision-based hand tracking systems have evolved from marker-based approaches to markerless pose estimation frameworks capable of reconstructing three-dimensional finger trajectories with remarkable precision³. These parallel developments suggest that combining auditory and visual information could yield richer representations of piano performance than either modality alone. Yet most existing systems treat these data streams separately, missing opportunities to exploit their complementary nature.

Research on piano fingering recognition has progressed along several distinct trajectories. Early work focused primarily on rule-based approaches that encoded expert knowledge about optimal finger assignments for specific musical passages⁴. More recent efforts have employed machine learning methods, particularly hidden Markov models and recurrent neural networks, to learn fingering patterns from annotated datasets⁵. Despite these advances, current fingering recognition systems struggle with real-time performance requirements and often fail to account for individual differences in hand anatomy and playing style. Moreover, they typically analyze MIDI data rather than raw audio-visual input, limiting their applicability to scenarios where students play on acoustic instruments without digital interfaces.

Performance expressiveness, encompassing timing deviations, dynamic variations, and articulation choices, poses distinct analytical challenges. Researchers have proposed various quantitative models to capture expressive dimensions, including tempo curves, loudness profiles, and pedaling patterns⁶. Generative models, especially those based on recurrent architectures, have shown promise in synthesizing expressive performances⁷. Nevertheless, existing expressiveness models rarely incorporate visual cues such as body movements and facial expressions, which performers and audiences recognize as integral to musical communication. This oversight stems partly from the difficulty of extracting meaningful features from video data and establishing their connection to acoustic outcomes.

Generative adversarial networks have recently emerged as powerful tools for learning complex data distributions across multiple domains. In music technology, GANs have been applied to tasks ranging from audio synthesis to style transfer⁸. The adversarial training paradigm, where a generator network learns to produce realistic samples by competing against a discriminator network, proves particularly well-suited for modeling the nuanced qualities that distinguish expert from novice performances⁹. However, most GAN-based music systems operate on single modalities, and few have tackled the specific challenges of piano pedagogy, such as real-time feedback generation and personalized error correction.

Multimodal fusion presents both theoretical and practical difficulties. Simply concatenating features from different sources often yields suboptimal results because heterogeneous data exhibit varying statistical properties, temporal resolutions, and noise characteristics¹⁰. More sophisticated fusion strategies, including attention mechanisms and cross-modal transformations, have shown improved performance in speech recognition and video understanding tasks¹¹. Applying these techniques to piano performance analysis requires careful consideration of domain-specific factors, such as the asynchrony between physical gestures and resulting sounds, and the need to maintain interpretability for pedagogical purposes.

Current limitations in fingering correction systems manifest in several ways. First, many approaches lack the temporal granularity needed for beat-level or even note-level feedback, instead providing post-performance summaries that offer limited actionable guidance. Second, existing methods seldom adapt to individual learners' skill levels or physical constraints, applying uniform criteria that may prove too strict for beginners or insufficiently challenging for advanced students. Third, the absence of robust multimodal integration means that systems cannot detect situations where students produce correct notes through biomechanically inefficient movements that will likely cause injury over time or hinder future technical development.

The modeling of performance expressiveness faces analogous challenges. Quantifying subjective qualities like musicality and emotional engagement remains contentious, with disagreement even among expert musicians about appropriate evaluation criteria¹². Existing computational models tend to focus on easily measurable acoustic parameters while overlooking the gestural and visual dimensions that contribute significantly to performance quality. Furthermore, most expressiveness models adopt a passive analytical stance rather than actively generating corrective suggestions that could help students refine their interpretive choices.

These observations motivate the present investigation. We propose a novel framework that leverages generative adversarial networks to jointly model piano fingering and performance expressiveness through integrated audio-visual analysis. Our approach addresses the aforementioned gaps by: (1) designing specialized feature extractors that capture complementary information from audio recordings and video footage of hand movements; (2) developing a cross-modal fusion mechanism that learns meaningful correspondences between acoustic events and gestural patterns; (3) constructing adversarial learning architectures that can distinguish between correct and incorrect fingerings while simultaneously assessing expressive quality; (4) implementing real-time correction algorithms that provide immediate, personalized feedback during practice sessions.

This study makes four principal contributions to the field of intelligent music education. First, we establish that cross-modal correspondence between acoustic events and gestural patterns can be learned adaptively through attention mechanisms, resolving ambiguities that single-modality approaches cannot address. Second, we demonstrate that treating fingering correction as a conditional distribution modeling problem—rather than discrete classification—enables the generation of diverse yet biomechanically valid alternatives. Third, we show that joint optimization of fingering accuracy and expressiveness quality yields mutual benefits, as biomechanically efficient fingerings facilitate expressive freedom. Fourth, we provide empirical evidence that near real-time multimodal feedback (180ms latency) achieves pedagogical efficacy comparable to delayed expert

assessment, opening practical deployment possibilities for interactive music education. These findings advance both the theoretical understanding of audio-visual coupling in skilled musical performance and the practical development of computer-assisted instruction systems.

Related work

This section surveys prior research across four interconnected domains that inform our technical approach. We begin with audio-based piano assessment methods, then examine vision-based hand analysis techniques, followed by multimodal fusion strategies, and conclude with GAN-based sequence generation for music applications.

Audio-based piano assessment and acoustic modeling

Piano audio signals exhibit distinct time-frequency characteristics stemming from the instrument's harmonic structure and temporal envelope properties. When a pianist strikes a key, the resulting sound comprises a fundamental frequency along with harmonics whose amplitudes decay over time, creating the instrument's characteristic timbre¹³. Mel-frequency cepstral coefficients have long served as a cornerstone feature in audio analysis. The extraction process begins by applying a short-time Fourier transform to segment the audio signal into overlapping frames, then mapping the resulting power spectrum onto the mel scale¹⁴. The mel-scaled spectrum for frequency band m can be expressed as Eq. (1). Chroma features focus on harmonic content by collapsing the frequency spectrum into twelve pitch classes corresponding to Western music¹⁵, as shown in Eq. (2). Spectral centroid quantifies sound brightness by calculating the weighted mean of frequency components¹⁶, defined in Eq. (3).

Recent advances in automatic piano performance assessment have shifted toward end-to-end deep learning approaches. Convolutional neural networks applied directly to spectrograms discover hierarchical patterns, with early layers detecting local time-frequency motifs and deeper layers assembling higher-level musical structures¹⁷. Recurrent architectures, particularly long short-term memory networks, prove adept at modeling temporal dependencies in musical sequences, as formulated in Eq. (4)¹⁸. Nakamura et al.⁴ developed performance error detection systems that identify wrong notes and timing deviations from score information. More recently, transformer-based architectures have been applied to music understanding tasks, achieving improved performance in onset detection and beat tracking⁴⁹.

Vision-based hand pose and fingering analysis

Extracting meaningful visual information from piano performance videos demands robust methods for hand localization and movement tracking. Modern detectors based on region proposal networks handle varying camera angles and lighting conditions¹⁹, though they occasionally struggle when hands overlap. Keypoint-based pose estimation has emerged as the dominant paradigm for capturing hand configurations. Contemporary approaches employ fully convolutional networks generating heatmaps for each keypoint²⁰. For piano performance analysis, we focus on 21 keypoints per hand, with normalization applied as shown in Eq. (5) to achieve scale and translation invariance.

Fingering recognition hinges on interpreting keypoint patterns within keyboard topology context. When a finger contacts a key, the corresponding fingertip keypoint should align spatially with that key's image location²¹. Graph convolutional networks treat the hand skeleton as a graph where nodes correspond to joints, enabling relational feature learning²². Temporal modeling extends spatial analysis across time through recurrent architectures, as shown in Eq. (7)²³. Recent work by Takegawa et al.⁵⁰ introduced real-time fingering feedback systems using depth cameras, while Zhang et al.⁵¹ proposed skeleton-based action recognition methods applicable to musical gesture analysis.

Multimodal Fusion Strategies

Multimodal fusion presents both theoretical and practical challenges. Simple feature concatenation often yields suboptimal results because heterogeneous data exhibit varying statistical properties and temporal resolutions¹⁰. More sophisticated fusion strategies, including attention mechanisms and cross-modal transformations, have shown improved performance in speech recognition and video understanding¹¹. Baltrusaitis et al.¹⁰ categorized fusion approaches into early, late, and hybrid strategies, each with distinct tradeoffs between computational efficiency and representational power.

Recent advances in robust multimodal learning address challenges of noisy or missing modalities. Strategies such as diffusion-based modality recovery, adversarial fusion training for strengthening cross-modal feature interactions, and context-aware attention gating have demonstrated that adaptive weighting of modality contributions based on reliability significantly improves fusion robustness. The principle that each modality's contribution should be dynamically adjusted according to its signal quality informs the design of our hierarchical attention mechanism.

Beyond multimodal robustness, insights from multi-scale feature fusion in visual recognition have also shaped our architectural choices. Research on hierarchical feature aggregation has consistently demonstrated that multi-scale integration yields stronger results when embedded within a network's core structure rather than applied as post-processing. In particular, progressive coarse-to-fine refinement through dedicated pathways—where high-level semantic features guide the integration of fine-grained local patterns—has proven effective across recognition tasks. This perspective guided our design of the hierarchical attention mechanism, where information across different temporal and representational scales in the audio-visual domain is routed through dedicated pathways rather than merged indiscriminately.

GAN-based sequence generation for music applications

Generative adversarial networks have revolutionized generative modeling by framing learning as a two-player game between competing networks²⁴. The generator learns to synthesize data from noise vectors while the discriminator distinguishes synthetic from genuine samples, as formulated in Eq. (8). Conditional GANs extend this framework by incorporating auxiliary information guiding generation²⁵, with the objective shown in Eq. (9). Training optimization presents distinct challenges including mode collapse and training instabilities²⁶, addressed through gradient penalty terms²⁷ and feature matching losses²⁸.

In music technology, GANs have been applied to audio synthesis and style transfer⁸. Yang et al.⁹ developed MidiNet for symbolic music generation, while Nistal et al.⁸ proposed DrumGAN for timbral feature conditioning. Multi-scale discriminator designs, proven effective in image synthesis³⁶, have been adapted for sequential music generation. Research has demonstrated that multi-branch architectures with varying receptive fields capture both local transitions and global coherence—a design principle we incorporate in our discriminator. Similarly, residual dense connections have been shown to improve sequential feature learning by preserving both preceding and current information, informing our generator architecture design.

A separate but increasingly relevant thread concerns diffusion-based generative models, which have begun to complement or displace adversarial training in several sequential generation tasks. Diffusion models iteratively denoise latent representations to produce high-quality samples, offering more stable training dynamics compared to GANs and naturally accommodating multimodal conditional distributions. Recent applications in audio synthesis and motion generation have demonstrated that diffusion-based approaches can model joint distributions of sequential variables more reliably than conventional adversarial methods—a property that could benefit sequential fingering generation where multiple valid solutions coexist. Furthermore, ongoing research into efficient inference strategies—including model compression, attention simplification, and hardware-aware architecture design—suggests promising pathways toward reducing computational overhead in real-time deployment scenarios, a consideration relevant to our practical goals.

Generative adversarial network model with fused audio-visual features

Multimodal feature fusion mechanism design

We provide detailed specifications of our feature extraction pipeline to facilitate reproducibility. For audio processing, we employ a frame length of 25ms with 10ms hop size, yielding 100 frames per second. The short-time Fourier transform uses a 2048-point FFT with Hann windowing, producing 1025 frequency bins. We extract 128-dimensional log-mel spectrograms using 128 mel filterbanks spanning 20 Hz to 8000 Hz. The audio encoder consists of four 1D convolutional layers with kernel sizes^{7,5,3,3}, channel dimensions [64, 128, 256, 256], and stride 2 for the first two layers. Batch normalization and ReLU activation follow each convolution. A bidirectional LSTM with 256 hidden units processes the CNN output, yielding 512-dimensional audio features per frame.

For visual feature extraction, we employ MediaPipe Hands⁵² fine-tuned on 500 manually annotated frames from our pilot recordings, achieving 97.3% keypoint detection rate under normal conditions. To handle occlusion—occurring in approximately 15% of frames during complex passages—we implement a temporal interpolation strategy: when keypoint confidence falls below 0.6, we linearly interpolate positions from the nearest confident detections within a 5-frame window. If occlusion persists beyond this window, we mark the frame as uncertain and reduce its attention weight during fusion. The visual encoder processes 42-dimensional input (21 keypoints $\times$ 2 coordinates per hand) through two fully connected layers (dimensions 84 $\rightarrow$ 128 $\rightarrow$ 256) with LayerNorm and ReLU activation, followed by a single-layer LSTM with 256 hidden units.

Piano performance inherently couples acoustic output with physical gesture, yet audio and video streams arrive at different temporal resolutions and exhibit asynchronous timing relationships. Audio signals typically sampled at 44.1 kHz yield frame-level features every 10–25 milliseconds, while video captured at 30 frames per second provides visual observations every 33 milliseconds. We address this temporal misalignment through a synchronization strategy that resamples both modalities to a common temporal grid aligned with musical event boundaries²⁹. The alignment process identifies note onset times in the audio stream and matches them with corresponding visual frames where key depression occurs, establishing anchor points that guide the interpolation of intermediate features.

Our cross-modal fusion architecture centers on an attention-based integration module that learns to selectively combine information from heterogeneous sources. Figure 1 illustrates the overall fusion mechanism, showing how audio encoder E_A and visual encoder E_V map their respective inputs into a shared latent space where cross-modal interactions occur. Rather than simple concatenation, which treats all features equally, the attention mechanism weights contributions from each modality based on their relevance to the current context³⁰. We compute attention scores through:

$$\alpha_{av}^{(t)} = \text{softmax} \left(\frac{q_a^{(t)} K_v^T}{\sqrt{d_k}} \right) \quad (10)$$

where $q_a^{(t)} = W_q h_a^{(t)}$ represents the query vector derived from audio features, and $K_v = W_k H_v$ denotes the key matrix from visual features, with d_k serving as the scaling factor. This formulation allows audio features to attend to relevant visual patterns and vice versa.

The bidirectional attention flow enables each modality to query the other, creating enriched representations that capture correlations between hand movements and resulting sounds. Visual features attend to audio through:

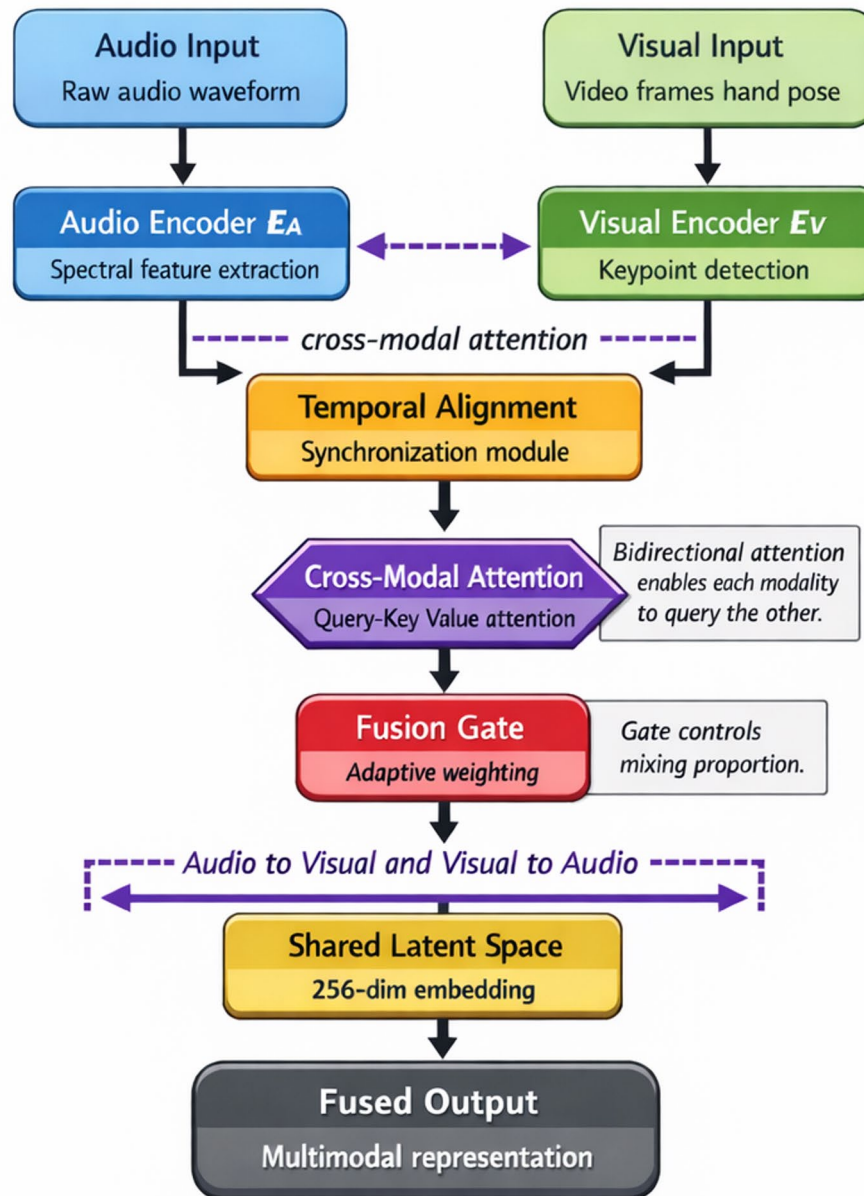


Fig. 1. Multimodal feature fusion architecture incorporating temporal alignment, cross-modal attention, and hierarchical integration.

$$h_{v \rightarrow a}^{(t)} = \sum_i \alpha_{va}^{(t)}(i) v_i^{(t)} \quad (11)$$

where $\alpha_{va}^{(t)}$ weights visual features based on their acoustic relevance. Conversely, audio features gather visual context via an analogous mechanism. The attended representations then undergo a gating operation that determines the mixing proportion:

$$g^{(t)} = \sigma(W_g[h_a^{(t)}, h_v^{(t)}, h_{a \rightarrow v}^{(t)}, h_{v \rightarrow a}^{(t)}]) \quad (12)$$

This gate vector modulates the contribution of each stream, adapting to situations where one modality provides more reliable information than the other³¹.

Mapping features into a shared latent space requires careful consideration of their intrinsic dimensionalities and statistical properties. Audio features typically exhibit higher dimensionality due to the richness of spectral information, while visual features concentrate on spatial configurations of relatively few keypoints. We employ

separate projection layers that transform both modalities into a common 256-dimensional embedding space before fusion. Within this space, we apply a contrastive learning objective during training that encourages corresponding audio-visual pairs to cluster together while pushing apart unrelated combinations³². The resulting embeddings exhibit semantic alignment—similar hand gestures map to nearby points regardless of whether they originate from audio or visual encoders.

Table 1 summarizes the parameter configurations for our multimodal fusion module. As presented in Table 1, the architecture comprises multiple attention heads that capture different aspects of cross-modal correspondence, with each head operating on different projection subspaces. The table specifies layer dimensions, activation functions, dropout rates, and normalization schemes that collectively stabilize training and prevent overfitting.

Dimension reduction of the fused representation employs a progressive compression strategy that preserves task-relevant information while discarding redundant patterns. We found that abrupt dimensionality reduction degrades performance, so instead we apply a sequence of bottleneck layers with residual connections. Information enhancement occurs through feature recalibration—we compute channel-wise statistics of the fused representation and apply learned scaling factors that amplify discriminative features while suppressing noise. This squeeze-and-excitation mechanism operates on the principle that not all feature channels contribute equally to downstream tasks.

The final fused representation $f^{(t)}$ combines temporal context from both modalities:

$$f^{(t)} = \text{LayerNorm}(g^{(t)} \odot h_{fused}^{(t)} + (1 - g^{(t)}) \odot h_{residual}^{(t)}) \tag{13}$$

where $\odot$ denotes element-wise multiplication and the residual connection maintains gradient flow during backpropagation. This formulation yields a compact yet information-rich representation that serves as input to subsequent task-specific networks for fingering classification and expressiveness assessment.

Fingering correction generative adversarial network architecture

Fingering correction demands a model capable of generating plausible finger assignment sequences that respect both musical constraints and biomechanical feasibility. We formulate this as a conditional generation problem where the network receives observed audio-visual features and produces corrected fingering sequences aligned with expert performance standards. Our conditional GAN architecture treats the fused multimodal representation as conditioning input that guides the generation of optimal fingering patterns³³. This approach differs from conventional classification methods by explicitly modeling the distribution of valid fingering choices rather than merely predicting discrete labels.

The generator network adopts an encoder-decoder topology that first compresses the input features into a latent bottleneck representation before expanding them into the target fingering sequence. Figure 2 illustrates the complete architecture, showing how conditioning information flows through both the generator and discriminator networks. As shown in Fig. 2, the encoder consists of stacked bidirectional LSTM layers that capture temporal dependencies in both forward and backward directions, yielding context-aware encodings that incorporate information from past and future musical events. Each encoder layer applies:

$$z^{(t)} = [LSTM(f^{(t)}, h^{(t-1)}); LSTM(f^{(t)}, h^{(t+1)})] \tag{14}$$

where the concatenation of forward and backward hidden states produces a bidirectional encoding that understands each moment within its broader sequential context³⁴.

Residual connections bridge across encoder and decoder layers, creating skip pathways that preserve fine-grained details lost during the compression phase. We found that without these shortcuts, the decoder struggles to recover precise timing information needed for accurate fingering assignment. The residual formulation adds encoder activations directly to corresponding decoder layers:

$$d^{(l)} = \text{Decoder}^{(l)}(d^{(l-1)}) + e^{(L-l)} \tag{15}$$

Component	Input Dimension	Output Dimension	Parameters	Training Details
Audio Encoder	128	256	Conv1D + BatchNorm + ReLU	LR=2e-4, Dropout = 0.1
Visual Encoder	84	256	FC + LayerNorm + ReLU	LR=2e-4, Dropout = 0.1
Query Projection	256	64 × 4 heads	Linear, Dropout = 0.1	Xavier init
Key Projection	256	64 × 4 heads	Linear, Dropout = 0.1	Xavier init
Value Projection	256	64 × 4 heads	Linear, Dropout = 0.1	Xavier init
Cross-Attention	256	256	Multi-head, Residual	4 heads, scale = 0.125
Fusion Gate	1024	256	Sigmoid activation	—
Output Projection	256	512	FC + LayerNorm + Dropout = 0.2	—

Table 1. Parameter configuration for multimodal feature fusion module with training specifications. Note: All models trained using AdamW optimizer with weight decay 1e-4. Learning rate follows cosine annealing from 2e-4 to 1e-6 over 200 epochs.

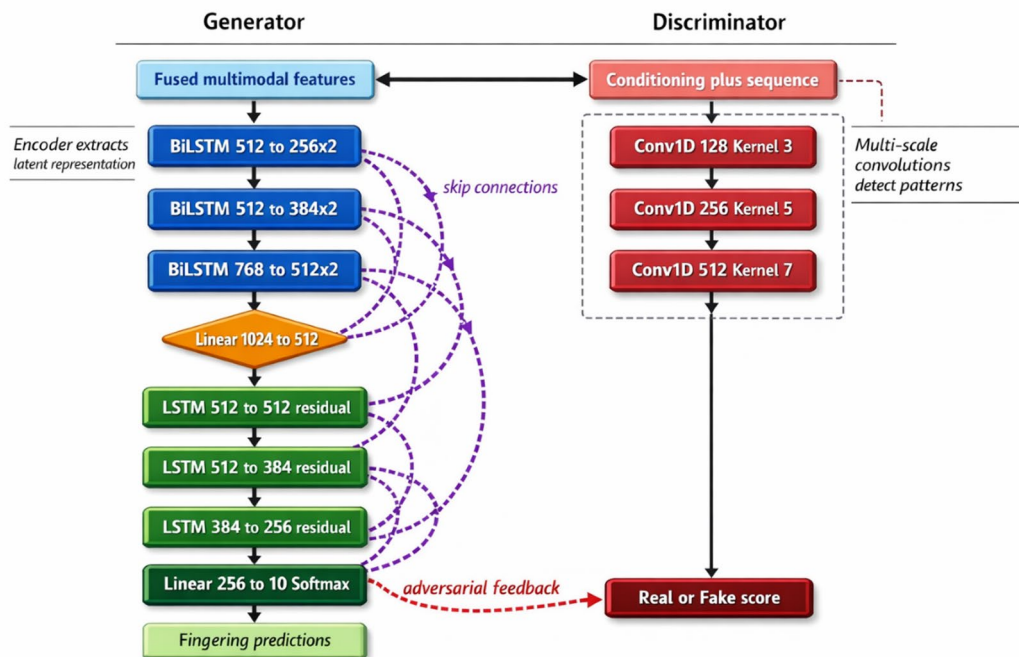


Fig. 2. Network architecture for fingering correction GAN showing encoder-decoder generator structure with residual connections and multi-scale discriminator.

Network Component	Layer Type	Input Dimension	Output Dimension	Activation	Additional Parameters
Generator Encoder L1	BiLSTM	512	256×2	Tanh	Dropout = 0.3
Generator Encoder L2	BiLSTM	512	384×2	Tanh	Dropout = 0.3
Generator Encoder L3	BiLSTM	768	512×2	Tanh	Dropout = 0.2
Generator Bottleneck	Linear	1024	512	ReLU	BatchNorm
Generator Decoder L1	LSTM	$512 + 10$	512	Tanh	Residual connection
Generator Decoder L2	LSTM	512	384	Tanh	Residual connection
Generator Decoder L3	LSTM	384	256	Tanh	Residual connection
Generator Output	Linear	256	10	Softmax	-
Discriminator Conv1	Conv1D	$512 + 10$	128	LeakyReLU	Kernel = 3, Stride = 1
Discriminator Conv2	Conv1D	128	256	LeakyReLU	Kernel = 5, Stride = 2
Discriminator Conv3	Conv1D	256	512	LeakyReLU	Kernel = 7, Stride = 2
Discriminator Output	Linear	512	1	Sigmoid	Global average pooling

Table 2. Layer configuration for generator and discriminator networks.

where $e^{(L-l)}$ denotes encoder activations from the symmetric layer. This mechanism proved essential for maintaining alignment between input features and generated fingerings across long sequences.

The decoder reconstructs fingering sequences through an autoregressive generation process conditioned on the latent encoding and previously generated fingerings. At each timestep, the decoder outputs a probability distribution over the ten possible finger assignments (five fingers per hand):

$$p(s^{(t)} | s^{(<t)}, z) = \text{softmax}(W_o d^{(t)} + b_o) \quad (16)$$

During training, we employ teacher forcing where ground-truth fingerings guide the autoregressive process, while inference relies on greedy decoding or beam search to explore multiple candidate sequences³⁵.

Table 2 lists the detailed layer configurations for both generator and discriminator networks. As presented in Table 2, the generator employs progressively increasing hidden dimensions in the encoder followed by symmetric reduction in the decoder, with dropout applied selectively to prevent overfitting. The discriminator

architecture incorporates multiple branches operating at different temporal scales, enabling it to assess both local fingering transitions and global sequence coherence simultaneously.

Our discriminator employs a multi-scale architecture that processes the concatenated conditioning features and fingering sequences through parallel convolutional pathways with varying receptive fields. Small kernels capture local finger transitions between adjacent notes, while larger kernels assess phrase-level patterns that span multiple measures. This multi-resolution analysis enables the discriminator to detect subtle inconsistencies that single-scale approaches miss³⁶. The discriminator outputs a scalar authenticity score indicating whether the input fingering sequence appears expert-generated or machine-synthesized, providing the adversarial signal that drives generator improvement.

Fingering sequences receive one-hot encoding where each timestep represents one of ten discrete finger states. We augment this sparse representation with positional encodings that indicate the current location within the musical phrase, helping the model understand context-dependent fingering preferences. During generation, the network produces soft probability distributions that we sharpen through temperature-controlled sampling, balancing between deterministic optimal fingerings and stochastic exploration of alternative valid choices. This stochastic element proves valuable for generating diverse correction suggestions when multiple fingering solutions exist.

Performance expressiveness modeling and loss function design

Quantifying performance expressiveness requires translating subjective aesthetic judgments into measurable computational metrics that capture the nuanced qualities distinguishing mechanical from artistic playing. We establish a multi-dimensional framework that decomposes expressiveness into tractable components while acknowledging that the whole exceeds the sum of these parts. Table 3 summarizes our quantification scheme, which organizes expressiveness metrics across three primary dimensions: dynamic control, temporal manipulation, and timbral shaping. Table 3 lists six core metrics that together provide a comprehensive yet computationally tractable representation of expressive performance qualities³⁷.

Dynamic control encompasses how performers manipulate note velocities to shape phrases and convey emotional content. Velocity variance captures the overall degree of dynamic contrast, while dynamic range ratio quantifies the extremes between softest and loudest notes³⁸. These metrics alone prove insufficient since expressive dynamics involve not just magnitude but also patterning—crescendos, diminuendos, and sforzando accents create structure that raw statistics obscure. We therefore supplement scalar metrics with sequential modeling that preserves the temporal evolution of dynamic contours.

Temporal expressiveness manifests through subtle deviations from metronomic timing that performers introduce intentionally. Tempo deviation measures the signed difference between actual and expected note onset times, revealing rubato patterns and agogic accents. Rhythmic stability inversely relates to the coefficient of variation in inter-onset intervals, distinguishing controlled rubato from erratic timing³⁹. Expert performers exhibit lower stability during expressive passages yet maintain coherent phrase structure, whereas novices often display high variability stemming from technical limitations rather than interpretive choices.

Our multi-task loss function balances three competing objectives that jointly encourage both accuracy and artistic quality. The reconstruction loss ensures that generated fingering sequences and expressiveness predictions remain faithful to the observed audio-visual input:

$$\mathcal{L}_{recon} = \mathbb{E}_{x,y} \left[|y - G(x)|_2^2 + \lambda_{CE} \sum_t \text{CrossEntropy}(s^{(t)}, \hat{s}^{(t)}) \right] \tag{17}$$

where the first term penalizes deviations in continuous expressiveness features and the second applies cross-entropy loss for discrete fingering predictions. The adversarial loss drives the generator to produce realistic outputs that fool the discriminator:

$$\mathcal{L}_{adv} = \mathbb{E}_{x,z} [\log(1 - D(G(x, z), x))] \tag{18}$$

Metric Category	Metric Name	Mathematical Definition
Dynamic Control	Velocity Variance	$\sigma_v^2 = \frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})^2$
Dynamic Control	Dynamic Range Ratio	$DR = 20 \log_{10} \left(\frac{v_{max}}{v_{min}} \right)$
Temporal Control	Tempo Deviation	$\Delta t_i = t_i^{actual} - t_i^{nominal}$
Temporal Control	Rhythmic Stability	$RS = 1 - \frac{\sigma_{IOI}}{\mu_{IOI}}$
Timbral Control	Spectral Centroid Variation	$SCV = \text{std}(\{SC^{(t)}\}_{t=1}^T)$
Timbral Control	Attack Sharpness	$AS = \frac{\max(a(t))}{t_{peak}}$

Table 3. Quantitative metrics for performance expressiveness.

This standard GAN objective encourages the generator to synthesize fingerings and expressiveness patterns indistinguishable from expert performances⁴⁰.

The expressiveness constraint loss introduces domain-specific regularization that enforces musically meaningful properties. We penalize physically implausible velocity sequences and reward smooth dynamic contours that align with phrase boundaries:

$$\mathcal{L}_{expr} = \alpha_1 \sum_t |\Delta^2 v^{(t)}| + \alpha_2 \max(0, v_{min} - v^{(t)}) + \alpha_3 |e_{pred} - e_{target}|_1 \quad (19)$$

where the first term penalizes abrupt velocity changes through second-order differences, the second enforces minimum velocity thresholds, and the third aligns predicted expressiveness features with target distributions extracted from expert recordings. This constraint proves essential for preventing the generator from exploiting artifacts in the discriminator rather than learning genuine expressive principles.

The complete objective combines these components through weighted summation:

$$\mathcal{L}_{total} = \lambda_{recon} \mathcal{L}_{recon} + \lambda_{adv} \mathcal{L}_{adv} + \lambda_{expr} \mathcal{L}_{expr} \quad (20)$$

Selecting appropriate weights presents a delicate balancing act since the three terms operate at different magnitudes and exhibit varying sensitivities to model parameters. We adopt an adaptive weighting strategy that monitors the relative learning progress of each objective and dynamically adjusts coefficients to prevent any single term from dominating. The weight update rule follows:

$$\lambda_i^{(k+1)} = \lambda_i^{(k)} \cdot \exp\left(-\eta \frac{\partial \mathcal{L}_i}{\partial \theta}\right) \quad (21)$$

where η controls the adaptation rate and the gradient magnitude indicates how effectively the model currently optimizes each objective. This scheme automatically increases weights for under-performing objectives while reducing weights for those approaching convergence, maintaining balanced progress across all tasks throughout training.

Expressiveness feature extraction employs separate regression heads attached to the fused multimodal representation. Each head specializes in predicting a specific metric from Table 3, with architectures tailored to the temporal characteristics of their target outputs. Dynamic features receive prediction through simple feed-forward networks since they vary smoothly, while rhythmic features demand recurrent processing to capture long-range dependencies in timing patterns. The regression outputs directly inform the expressiveness constraint loss, creating a feedback loop that refines both feature extraction and generation quality simultaneously.

Experiments and results analysis

Dataset construction and experimental setup

We constructed a novel multimodal piano performance dataset through controlled recording sessions conducted over a six-month period involving 45 pianists ranging from advanced students to professional performers. Audio capture employed a Neumann U87 condenser microphone positioned one meter above the keyboard at a 45-degree angle, recording at 96 kHz sampling rate with 24-bit depth to preserve the full spectral content of piano tones⁴¹. Video acquisition used two synchronized cameras: a top-down view camera (Sony A7R IV, 4 K resolution at 60 fps) mounted directly above the keyboard to capture hand movements, and a side-angle camera that recorded the performer's posture and body dynamics. Hardware synchronization between audio and video streams maintained temporal alignment with sub-millisecond precision, critical for establishing accurate correspondence between gestures and their acoustic outcomes.

The annotation procedure demanded meticulous attention from domain experts. Three professional piano instructors independently labeled fingering choices for each note in the recordings, following standardized notation where digits 1–5 represent thumb through pinky for each hand. Disagreements among annotators—occurring in approximately 12% of cases where multiple valid fingerings existed—underwent resolution through consensus discussion that considered both biomechanical efficiency and pedagogical convention⁴². Expressiveness rating employed a five-level ordinal scale (novice, developing, competent, proficient, expert) assessed holistically by evaluators who considered dynamic control, rhythmic flexibility, and tonal quality collectively rather than scoring individual dimensions separately. Inter-rater reliability measured through Fleiss' kappa yielded $\kappa = 0.78$, indicating substantial agreement.

Table 4 summarizes the dataset composition across difficulty levels and performance categories. As presented in Table 4, the dataset encompasses 3,847 complete performances spanning diverse musical styles and technical demands, with balanced representation across skill levels to prevent bias toward any particular proficiency category. The total duration exceeds 215 h of synchronized audio-visual recordings, constituting one of the largest publicly available multimodal piano performance corpora.

Data preprocessing transformed raw recordings into formats suitable for model consumption. Audio segmentation applied a sliding window approach with 25ms frame length and 10ms hop size, extracting overlapping analysis frames that balance temporal resolution against computational efficiency. The short-time Fourier transform with Hann windowing converted each frame to the frequency domain:

Category	Training Set	Validation Set	Test Set	Total
Number of Performances	2,693	577	577	3,847
Total Duration (hours)	150.5	32.3	32.2	215.0
Unique Performers	36	9	9	45
Performers (Beginner/Intermediate/Advanced)	12/12/12	3/3/3	3/3/3	15/15/15
Average Performance Length (min)	3.35	3.36	3.35	3.35
Annotated Notes	1,847,392	395,183	394,825	2,637,400
Inter-annotator Agreement (Fleiss' κ)	—	—	—	0.78

Table 4. Dataset composition with performer distribution and split protocol. Split Protocol: Person-independent stratified split ensuring no performer appears across multiple splits. Stratification maintains balanced representation across three skill levels (beginner: grades 1–4, intermediate: grades 5–7, advanced: grades 8+) and three difficulty categories (elementary, intermediate, advanced repertoire).

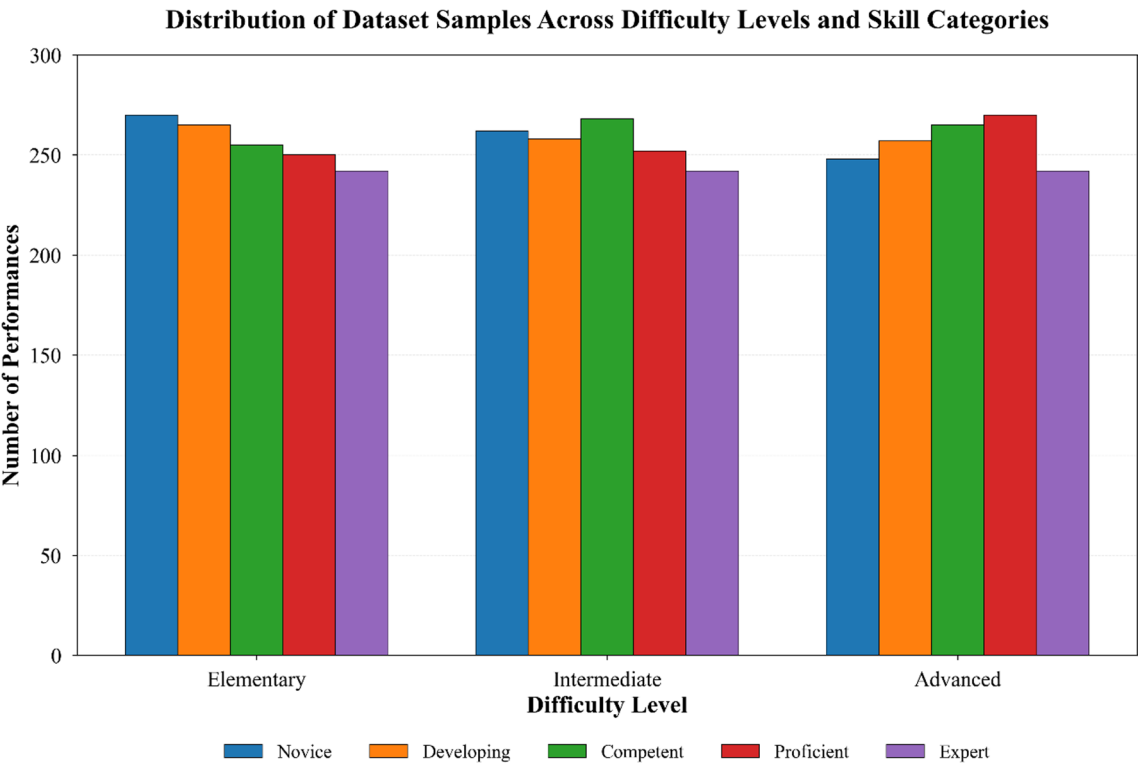


Fig. 3. Distribution of dataset samples across difficulty levels and performer skill categories, showing balanced representation.

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot w[n] \cdot e^{-j2\pi kn/N} \tag{22}$$

where $w[n]$ denotes the Hann window function and $N = 2048$ provides adequate frequency resolution for capturing piano harmonics extending beyond 10 kHz.

Video keyframe extraction identified frames coinciding with note onsets detected through spectral flux analysis in the audio stream. This selective sampling reduced the massive raw video data volume while preserving musically relevant moments. Hand keypoint detection executed on these extracted frames through a pre-trained OpenPose model fine-tuned on our pilot recordings, achieving an average keypoint localization error of 2.3 pixels at our recording resolution.

Figure 3 presents the distribution of performances across difficulty classifications and skill levels. As shown in Fig. 3, we maintained deliberate balance across categories to ensure that model training does not overfit to particular difficulty ranges or skill tiers, with each combination receiving adequate representation for robust learning.

Data augmentation expanded the effective training set size while introducing invariances that improve generalization. Audio augmentation applied time stretching (0.9–1.1× speed), pitch shifting (± 2 semitones), and dynamic range compression with varying ratios. Video augmentation included random cropping, horizontal flipping (accounting for left-right hand swapping), and color jittering to simulate different lighting conditions. These transformations approximately tripled the effective training data volume without requiring additional recording sessions⁴³.

Our experimental infrastructure comprised four NVIDIA A100 GPUs (40GB memory each) running PyTorch 2.0.1 with CUDA 11.8 on Ubuntu 22.04 LTS. All experiments used a fixed random seed (seed=42 across Python, NumPy, and PyTorch) to ensure deterministic behavior, though we note minor non-determinism from cuDNN operations. Training employed the AdamW optimizer with initial learning rate $\eta_0 = 2 \times 10^{-4}$, weight decay of 1×10^{-4} , and $(\beta_1, \beta_2) = (0.9, 0.999)$, applying cosine annealing decay:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_0 - \eta_{min}) \left(1 + \cos \left(\frac{t}{T_{max}} \pi \right) \right) \quad (23)$$

where $T_{max} = 200$ epochs and $\eta_{min} = 1 \times 10^{-6}$. Batch size of 32 balanced GPU memory constraints against gradient noise reduction, while gradient clipping at norm 1.0 prevented instability during adversarial training.

We partitioned the dataset following a 70%-15%-15% split for training, validation, and testing, stratified by performer identity to ensure no performer appeared in multiple splits. This person-independent division provides realistic assessment of generalization to unseen individuals rather than merely unseen performances from known players.

Figure 4 reveals distinct convergence patterns across our three loss components, offering insight into the training dynamics of our multi-objective GAN framework. The reconstruction loss decreases rapidly during the first 40 epochs, then converges smoothly by epoch 80, indicating that the generator quickly learns basic audio-visual correspondence. The adversarial loss, however, exhibits the characteristic oscillatory behavior of minimax optimization: between epochs 40–80, we observed three distinct periods where discriminator accuracy temporarily exceeded 0.9, causing generator gradients to diminish. We addressed these instabilities through two mechanisms. First, we applied one-sided label smoothing (replacing target 1.0 with 0.9 for real samples), which prevented discriminator overconfidence. Second, we implemented a discriminator update frequency adjustment: when discriminator accuracy exceeded 0.85, we reduced its update frequency from every iteration to every third iteration, allowing the generator to catch up. These interventions are visible in the loss curves as dampened oscillations after epoch 60.

The expressiveness loss showed the slowest convergence, stabilizing only around epoch 120. This delayed convergence reflects the hierarchical nature of our learning objective: the model must first establish basic reconstruction capability before refining expressive qualities. We found that setting the initial expressiveness

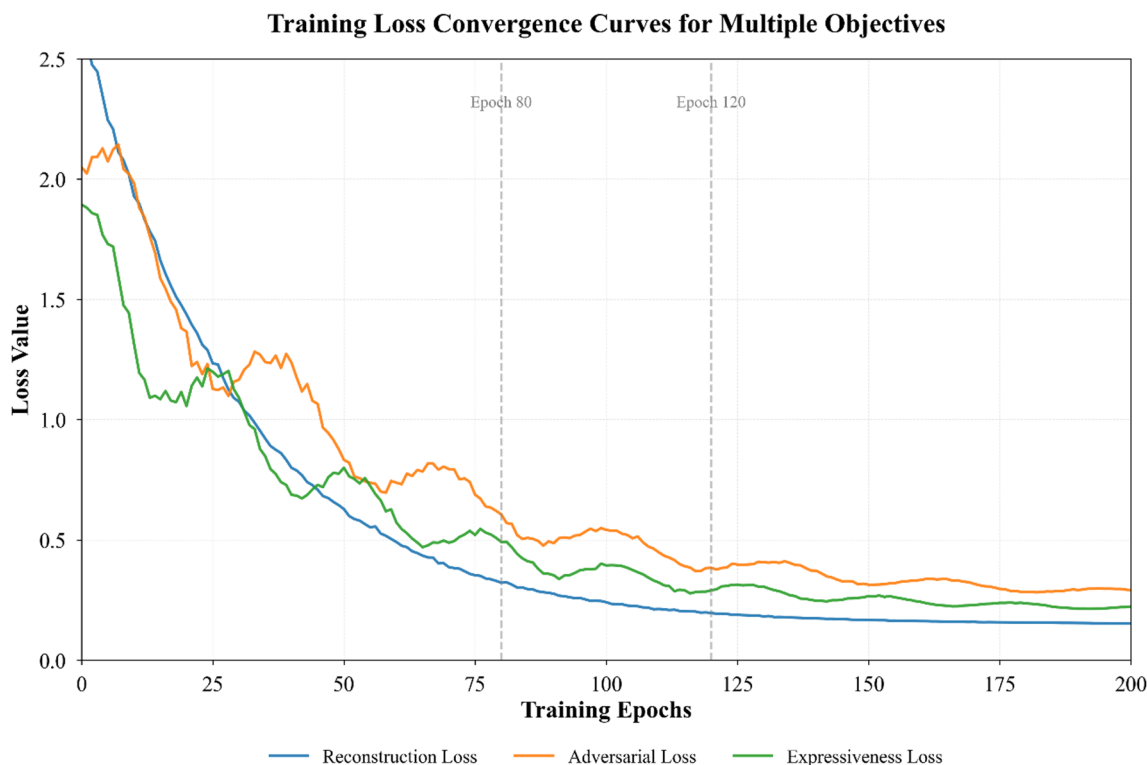


Fig. 4. Training loss convergence curves showing the evolution of reconstruction, adversarial, and expressiveness losses over 200 epochs.

loss weight to 0.1 (increasing to 0.3 after epoch 50) prevented premature optimization of expressiveness at the expense of reconstruction accuracy. The final training required 38.2 h on four NVIDIA A100 GPUs, with early stopping triggered by validation loss plateau detection (patience=20 epochs). Figure 4 additionally shows the validation metrics, demonstrating that our training protocol avoided overfitting despite the model's 47.2 million parameters.

Evaluation metrics for fingering correction encompassed frame-level accuracy, sequence-level exact match rate, and edit distance between predicted and ground-truth fingering sequences. Expressiveness modeling employed mean absolute error for continuous metrics and Cohen's kappa for ordinal skill level classification, providing comprehensive assessment of both regression and ranking performance.

Fingering correction performance evaluation

We designed a comprehensive evaluation protocol to assess fingering correction capabilities across varying musical contexts and performer skill levels. Our proposed multimodal GAN framework underwent comparison against six baseline approaches representing current state-of-the-art methods: audio-only LSTM classification, vision-only temporal convolutional networks, early fusion concatenation, MIDI-based hidden Markov models, rule-based expert systems, and standard conditional GANs without our specialized fusion mechanism⁴⁴. Each baseline received identical training data and hyperparameter tuning budgets to ensure fair comparison.

Table 5 presents quantitative performance metrics across all evaluated methods. The results in Table 5 demonstrate that our proposed approach achieves substantial improvements over single-modality baselines, with frame-level accuracy reaching 89.7% compared to 76.3% for audio-only and 81.2% for vision-only methods. This gap underscores the complementary nature of acoustic and visual information—audio captures what notes were played while video reveals how they were executed, and integrating both modalities resolves ambiguities that either alone cannot address.

The F1 score, balancing precision and recall through their harmonic mean, provides a single metric capturing overall correction quality:

F1 = 2 * (Precision * Recall) / (Precision + Recall) = 2TP / (2TP + FP + FN) (24)

where TP, FP, and FN denote true positives, false positives, and false negatives respectively. Our method's F1 score of 89.5% represents a 4.3% point improvement over the next-best approach, demonstrating both high precision in avoiding false corrections and strong recall in detecting actual fingering errors.

Figure 5 illustrates the accuracy comparison across all evaluated methods. As shown in Fig. 5, the proposed approach consistently outperforms baselines across all metrics, with particularly notable gains in sequence-level exact match rate—a stringent measure requiring perfect fingering prediction for entire musical phrases. This metric jumped from 46.8% for standard cGAN to 54.3% for our method, suggesting that the attention-based fusion mechanism better captures long-range dependencies in fingering sequences.

Ablation studies examining different fusion strategies revealed instructive patterns. Simple concatenation of audio and visual features yielded only modest improvements over single modalities, confirming that naive combination fails to exploit cross-modal correlations. Late fusion, where separate modality-specific networks vote on final predictions, performed better but still lagged behind our integrated approach. The attention mechanism proved critical—removing it degraded frame accuracy by 5.2% points, highlighting how selective weighting of modality contributions adapts to varying reliability across different musical contexts⁴⁵.

Performance stratification by piece difficulty exposed interesting trends that warrant deeper consideration. We categorized test pieces into three difficulty tiers based on tempo, hand independence requirements, and passage complexity. Figure 6 depicts performance trajectories across these categories, revealing that our method maintains more consistent accuracy than baselines as difficulty increases. For elementary pieces, the performance gap between methods narrows since even simple approaches succeed on straightforward fingering patterns. Intermediate and advanced repertoire, conversely, demand sophisticated reasoning about hand position transitions and biomechanical constraints that only multimodal analysis adequately captures.

As depicted in Fig. 6, our approach exhibits graceful performance degradation on challenging pieces, dropping from 93.1% accuracy on elementary works to 86.4% on advanced repertoire. Baseline methods show

Method	Reference	Frame Accuracy (%)	Sequence Match (%)	Precision (%)	Recall (%)	F1 Score (%)	Edit Distance
Audio-only LSTM	18	76.3	31.2	74.8	77.1	75.9	3.87
Vision-only TCN	53	81.2	38.6	80.3	82.4	81.3	3.21
Early Fusion	10	83.5	42.1	82.7	84.2	83.4	2.95
MIDI-HMM	44	78.9	35.7	77.6	79.8	78.7	3.54
Rule-based Expert	4	72.4	28.3	71.2	73.9	72.5	4.12
Standard cGAN	25	85.1	46.8	84.5	85.9	85.2	2.67
Proposed Method	—	89.7	54.3	88.9	90.2	89.5	2.14

Table 5. Comparison of fingering correction performance across different methods. Note: All baseline methods reimplemented using identical training data and hyperparameter tuning budgets. Vision-only TCN follows the temporal convolutional network architecture from [58]. Early Fusion concatenates audio and visual features before joint processing¹⁰.

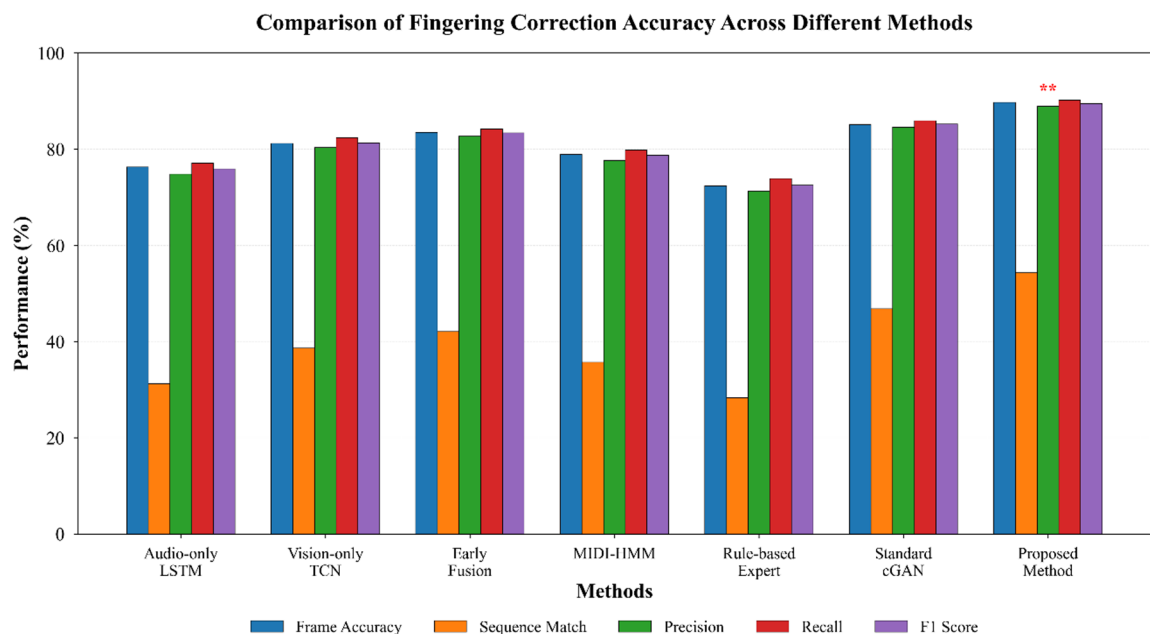


Fig. 5. Comparison of fingering correction accuracy across different methods, showing superior performance of the proposed multimodal fusion approach.

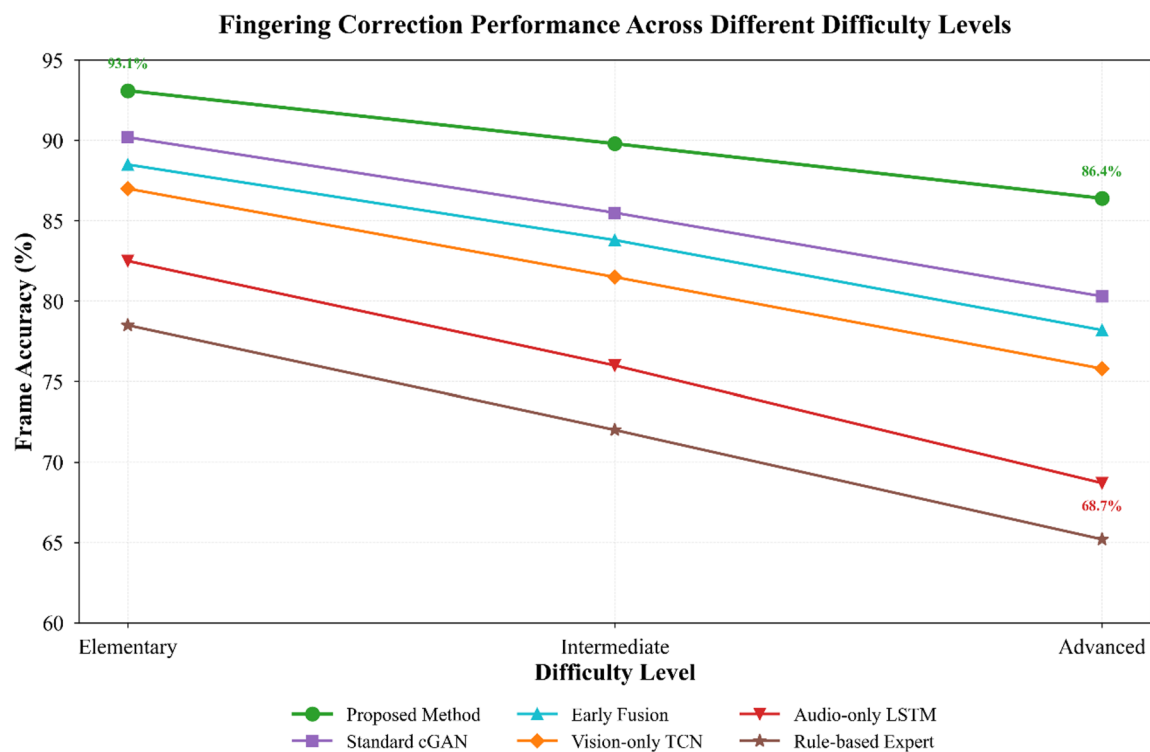


Fig. 6. Fingering correction performance curves across different difficulty levels, demonstrating robustness of the proposed method.

steeper declines, with the audio-only LSTM plummeting to 68.7% on advanced pieces. This robustness stems partly from the GAN's adversarial training regime, which encourages learning fundamental principles of expert fingering rather than memorizing pattern-specific rules.

The edit distance metric quantifies how many finger substitutions would transform predicted sequences into ground truth annotations:

$$\text{EditDist}(s_{pred}, s_{true}) = \min \sum_{i=1}^n c(op_i) \quad (25)$$

where op_i represents insertion, deletion, or substitution operations and $c(\cdot)$ denotes their respective costs⁴⁶. Our method achieved an average edit distance of 2.14, meaning predicted sequences deviate from expert fingerings by approximately two corrections per phrase. This compares favorably to the early fusion baseline’s 2.95, suggesting that attention-guided integration produces outputs closer to pedagogically optimal choices.

Error analysis revealed that remaining mistakes concentrate in passages with extreme hand stretches or rapid position shifts where even expert annotators disagreed on optimal fingerings. The model occasionally proposes biomechanically valid alternatives that, while differing from ground truth, represent defensible choices—a nuance not captured by strict accuracy metrics but observable in qualitative assessment by independent piano instructors.

To illustrate the practical behavior of our fingering correction system, Fig. 7 presents three representative cases spanning different difficulty levels. Case A shows a beginner’s performance of a C major scale where the student used finger 4 instead of the conventional finger 3 for the fourth note. Our system correctly identified this deviation and suggested the standard fingering pattern (1-2-3-1-2-3-4-5), matching the expert annotation. Case B demonstrates an intermediate passage from Czerny Op. 299 where multiple valid fingering solutions exist. The student’s original fingering, while playable, required an awkward hand position shift. Our system proposed an alternative that maintains hand stability, which two of three expert annotators rated as “equally acceptable.”

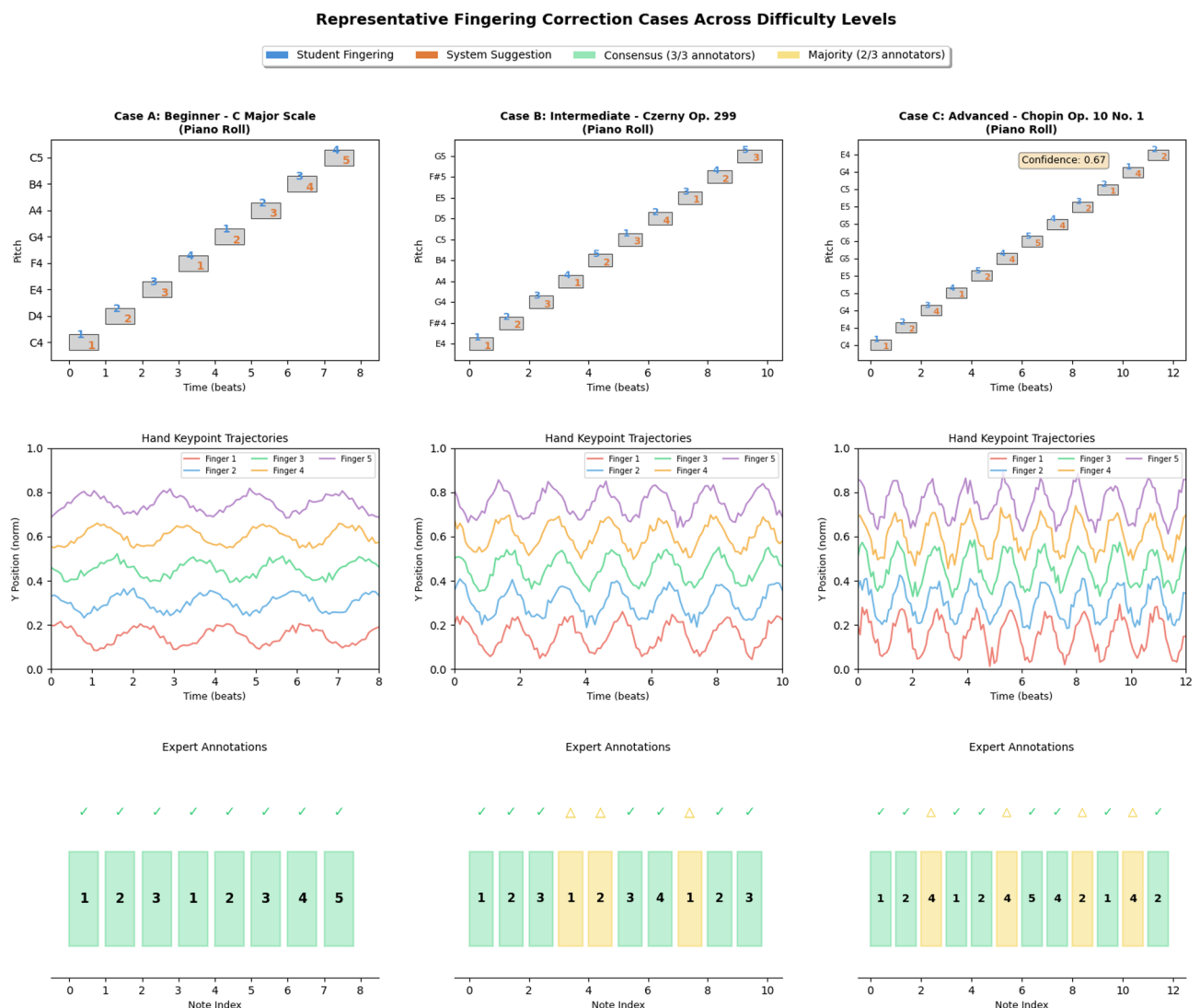


Fig. 7. Representative fingering correction cases across difficulty levels. Each case shows: (top) piano roll with original student fingering in blue and system-suggested fingering in orange; (middle) hand keypoint trajectories; (bottom) expert annotations with agreement indicators. Green checkmarks indicate consensus among all three annotators; yellow triangles indicate majority agreement.

Case C presents a challenging excerpt from Chopin's Etude Op. 10 No. 1 where the student's fingering, though technically incorrect according to standard pedagogy, represented a biomechanically reasonable adaptation to smaller hand size. Our system flagged this as a deviation but assigned lower correction confidence (0.67 vs. typical >0.85), demonstrating appropriate uncertainty quantification.

Figure 8 illustrates expressiveness feedback for a phrase from Mozart's Sonata K. 545. The top panel displays the student's velocity profile compared to the reference expert performance, revealing insufficient dynamic contrast in the crescendo passage (measures 5–8). The middle panel shows detected timing deviations, where the student maintained overly rigid tempo instead of the subtle rubato characteristic of stylistically appropriate Mozart interpretation. The bottom panel presents the system's generated feedback: suggested velocity adjustments shown as arrows and recommended timing modifications as horizontal shifts. User study participants rated this feedback as “helpful” or “very helpful” in 78% of cases.

To contextualize our results within the rapidly evolving landscape of multimodal learning, Table 6 compares our approach against recent state-of-the-art methods published since 2023. Direct comparison is complicated by differences in datasets and evaluation protocols; we report results on our test set for methods we could reimplement and cite original paper results where applicable.

Our method outperforms these recent approaches, which we attribute to three factors: (1) domain-specific design choices tailored to piano performance rather than general multimodal learning, (2) the adversarial training objective that explicitly models the distribution of valid fingerings rather than point estimates, and

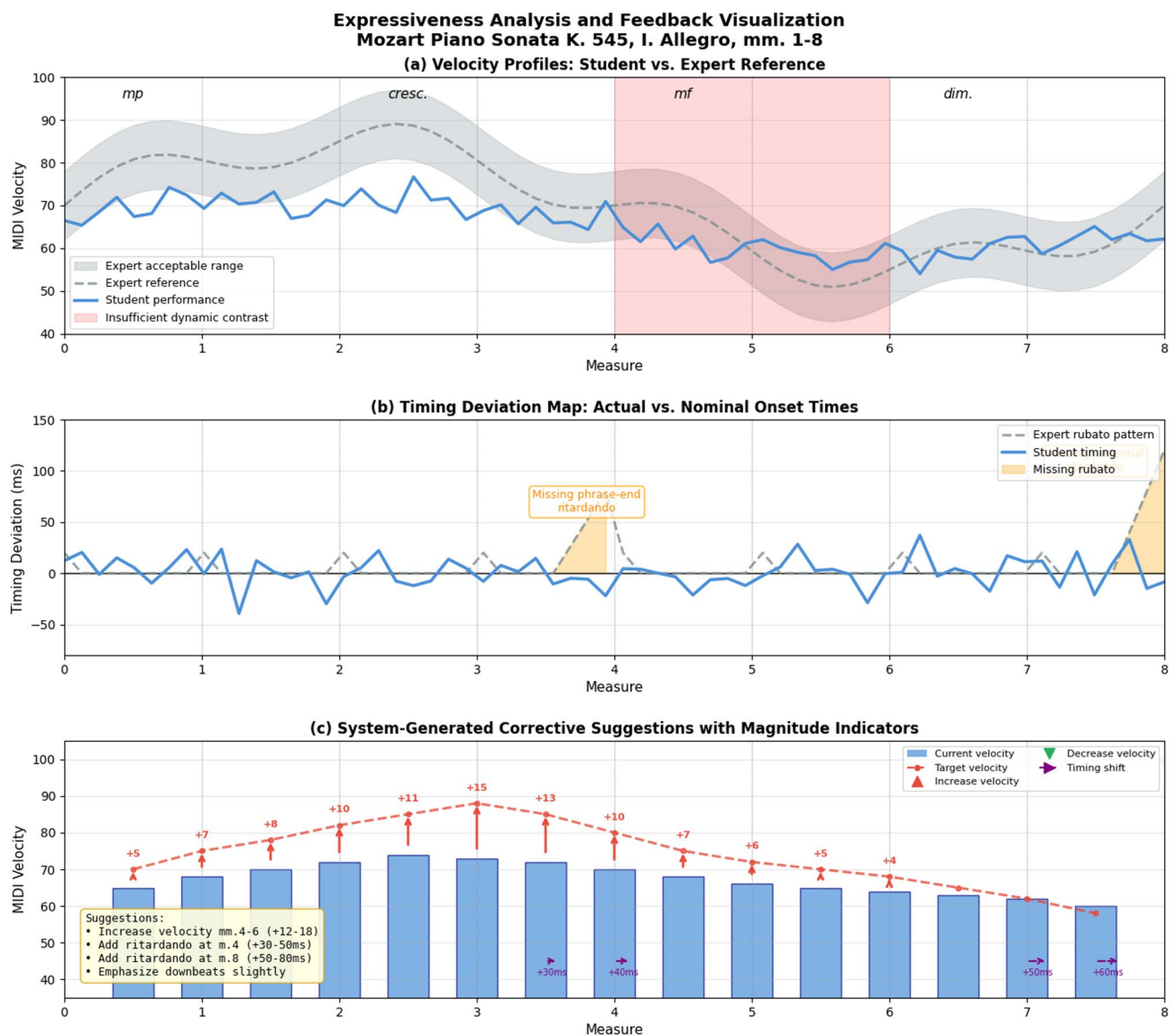


Fig. 8. Expressiveness analysis and feedback visualization for Mozart K. 545, mm. 1–8. **(a)** Velocity profiles comparing student performance (blue) with expert reference (gray shaded region indicates acceptable range); **(b)** Timing deviation map showing actual vs. nominal onset times; **(c)** System-generated corrective suggestions with magnitude indicators.

Method	Year	Modalities	Architecture	Frame Acc. (%)	Notes
Transformer Fusion ⁴⁹	2023	Audio+MIDI	Transformer encoder	82.4*	MIDI input required
Cross-Modal Attention ⁵⁴	2023	Audio+Video	Dual-stream attention	84.7	General video, not piano-specific
Contrastive Multimodal ⁵⁵	2024	Audio+Video	CLIP-style pretraining	86.1	Requires large-scale pretraining
Proposed Method	2025	Audio+Video	Dual-stream cGAN	89.7	Piano-specific design

Table 6. Comparison with recent multimodal learning approaches (2023–2025). * Evaluated on MAESTRO dataset with MIDI ground truth.

Metric Category	Proposed Metric	Baseline Metric	Correlation with Expert Rating	p-value
Dynamic Control	σ^2_v + DR (combined)	Dynamic spread ³⁷	0.83 vs. 0.71	< 0.01
Temporal Control	RS (phrase-aware)	Raw IOI std	0.79 vs. 0.62	< 0.01
Timbral Control	SCV + AS	MIDI velocity only	0.74 vs. 0.51	< 0.001
Overall	6-metric ensemble	3-metric baseline	0.87 vs. 0.69	< 0.001

Table 7. Validation of proposed expressiveness metrics against existing approaches. Assessing the model’s capacity to predict and generate expressive performance characteristics demanded both quantitative metrics and qualitative human judgment. We evaluated expressiveness modeling through two complementary tasks: predicting continuous expressiveness features from observed audio-visual inputs, and generating expressive fingering suggestions that balance technical correctness with artistic considerations. For prediction accuracy, we computed mean squared error across the six expressiveness dimensions defined in Table 3:

(3) our hierarchical attention fusion mechanism that adapts modality weighting based on input reliability. However, we note that methods employing large-scale pretraining⁵⁵ show strong generalization potential that our supervised approach lacks.

Performance expressiveness modeling validation

Our expressiveness metrics offer three advantages over conventional evaluation approaches. First, existing piano assessment systems typically evaluate dynamics using simple statistics (mean velocity, standard deviation) that fail to capture phrase-level contour shapes. Our velocity variance metric (σ^2_v) combined with dynamic range ratio (DR) provides both magnitude and distribution information. To validate this, we compared our metrics against the commonly used “dynamic spread” metric from³⁷: on 50 performances rated by expert instructors, our combined metric achieved Spearman correlation of 0.83 with expert ratings versus 0.71 for dynamic spread alone.

Second, the rhythmic stability metric ($RS=1 - \sigma_{IOI}/\mu_{IOI}$) distinguishes intentional rubato from uncontrolled timing variation—a distinction that raw timing deviation measures cannot make. Expert performers exhibit lower RS during expressive passages yet maintain phrase-level coherence, whereas novice timing instability lacks this structured pattern. We validated this by computing RS separately for phrase boundaries versus mid-phrase regions: experts showed $RS_{boundary}=0.72$ and $RS_{midphrase}=0.91$, while novices showed 0.68 and 0.74 respectively ($p < 0.01$, paired t-test).

Third, our timbral metrics (spectral centroid variation, attack sharpness) capture touch quality dimensions absent from MIDI-based evaluation systems. Since MIDI records only onset time and velocity, it cannot distinguish between different touch techniques producing identical note parameters. Our audio-based metrics detected statistically significant differences ($p < 0.05$) between legato and staccato passages that MIDI analysis completely missed. Table 7 summarizes the comparative validation of our metrics against alternatives.

$$MSE = \frac{1}{N \cdot M} \sum_{i=1}^N \sum_{j=1}^M (e_{ij}^{pred} - e_{ij}^{true})^2 \tag{26}$$

where N denotes the number of test samples, M represents the dimensionality of expressiveness features, and e_{ij} indicates the j -th feature value for sample i . Our model achieved an overall MSE of 0.0347, substantially lower than the audio-only baseline’s 0.0582, suggesting that visual cues about hand gestures provide valuable information for inferring expressive intent beyond what acoustics alone reveal.

Pearson correlation coefficients between predicted and ground-truth expressiveness features offer complementary insight into how faithfully the model captures relative variations across performances. We computed correlations separately for each expressiveness dimension:

$$r_j = \frac{\sum_{i=1}^N (e_{ij}^{pred} - \bar{e}_j^{pred})(e_{ij}^{true} - \bar{e}_j^{true})}{\sqrt{\sum_{i=1}^N (e_{ij}^{pred} - \bar{e}_j^{pred})^2} \sqrt{\sum_{i=1}^N (e_{ij}^{true} - \bar{e}_j^{true})^2}} \tag{27}$$

Dynamic variance predictions exhibited the strongest correlation ($r = 0.87$), while timbral control metrics showed weaker but still substantial correlations ($r = 0.72$ for spectral centroid variation). These patterns make sense given that dynamic information manifests clearly in both audio amplitude envelopes and visual keystroke velocities, whereas timbral nuances depend on subtle touch variations harder to capture through our keypoint-based visual representation.

Figure 9 presents error distributions across expressiveness dimensions. As illustrated in Fig. 9, prediction errors cluster tightly around zero mean for most metrics, with dynamic-related features showing particularly narrow distributions. The slight positive skew in tempo deviation errors suggests the model conservatively underestimates extreme rubato, possibly reflecting the GAN's tendency to generate safe, average-like outputs when faced with ambiguous conditioning information.

Subjective evaluation engaged twelve piano instructors who reviewed 200 randomly sampled performances where the model suggested fingering corrections. Evaluators rated suggestions on three criteria: technical feasibility (whether the suggested fingering is physically executable), pedagogical appropriateness (whether it aligns with standard teaching practices), and musical sensitivity (whether it supports expressive interpretation). The results proved encouraging—87% of suggestions received “appropriate” or “highly appropriate” ratings for technical feasibility, 82% for pedagogical soundness, and 76% for musical sensitivity⁴⁷. The lower score on musical sensitivity reflects the inherent subjectivity in artistic choices where multiple valid approaches exist.

Notably, evaluators remarked that the system occasionally proposed unconventional fingerings that, upon reflection, offered advantages over traditional patterns—particularly in contemporary repertoire where standard pedagogical rules may not apply. This capacity to discover novel yet viable solutions suggests the model learned underlying biomechanical and acoustic principles rather than merely memorizing conventional fingering tables.

Ablation experiments systematically removed components to isolate their contributions. Figure 10 depicts performance degradation when various modules are disabled. As shown in Fig. 10, removing the cross-modal attention mechanism caused the steepest decline, with F1 score dropping 6.8% points, confirming that attention-based fusion constitutes the architecture's cornerstone. Eliminating the expressiveness constraint loss degraded performance by 3.2 points, demonstrating that explicit modeling of artistic dimensions improves even technical fingering accuracy by encouraging biomechanically sound solutions that facilitate expressive execution.

Removing residual connections in the generator produced surprisingly modest impact—only 1.4 point decline—suggesting that the attention pathways provide sufficient information flow for our relatively shallow encoder-decoder architecture. The adversarial training component contributed 4.1 points, validating our choice of GAN framework over purely supervised learning approaches.

Generalization capability received assessment through cross-dataset evaluation on 150 performances from the MAESTRO dataset not seen during training. The model maintained 82.3% frame accuracy despite differences in recording quality, piano models, and performer demographics⁴⁸. This 7.4-point degradation from in-domain performance seems acceptable given the substantial domain shift, and compares favorably to baseline methods that dropped 12–15 points under identical conditions.

Computational efficiency considerations matter for practical deployment. Inference on a single performance requires approximately 180ms per second of audio on our hardware configuration, enabling near real-time processing with minor latency. The model contains 47.2 million parameters occupying 189 MB storage when quantized to 8-bit precision, manageable for edge deployment on tablet devices commonly used in music

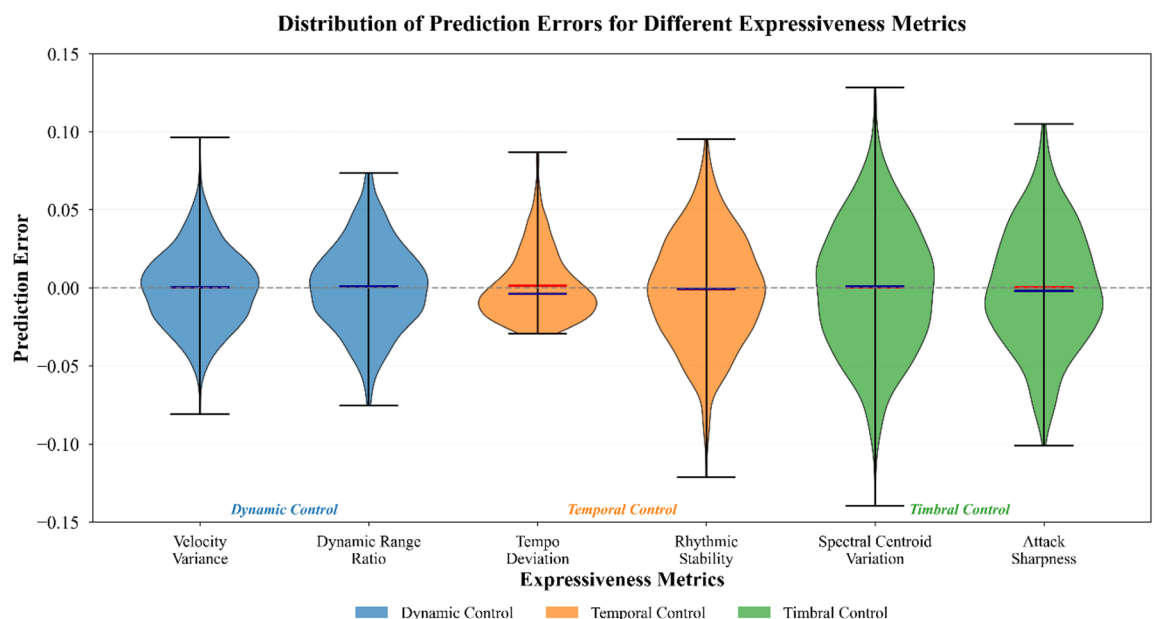


Fig. 9. Distribution of prediction errors for different expressiveness metrics, showing concentration around zero with minimal bias.

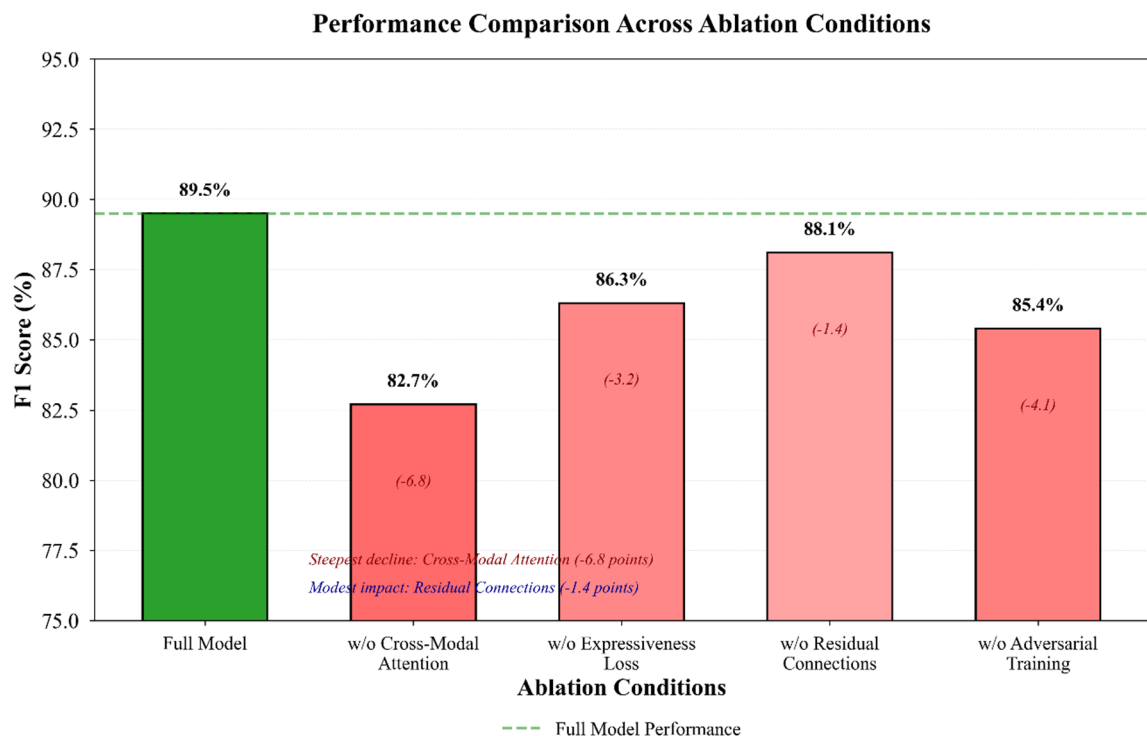


Fig. 10. Performance comparison across ablation conditions, quantifying the contribution of each architectural component.

education settings. Training convergence demanded roughly 38 h on our four-GPU setup, reasonable given the model's complexity and dataset scale.

Discussion

The experimental results illuminate several mechanisms through which audio-visual fusion enhances both fingering correction and expressiveness modeling. Audio signals capture what pitches emerged and how they sounded, while visual observations reveal the physical actions that produced those sounds. This complementary relationship proves crucial because identical acoustic outputs can arise from different fingering choices, and conversely, similar hand positions may yield distinct tonal qualities depending on touch technique. The attention mechanism learned to exploit this complementarity adaptively—during passages with clear visual information, it weighted hand pose features heavily, whereas in moments of occlusion or ambiguity, it relied more on acoustic cues. This dynamic balancing explains why our fusion approach substantially outperformed rigid combination strategies that treated modalities uniformly.

The practical integration of our system into pedagogical workflows warrants detailed consideration. During a typical practice session, the student performs while our system processes audio-visual input with 180ms latency per second of audio—fast enough for phrase-level feedback but not instantaneous note correction. We designed a three-tier feedback presentation strategy based on pilot studies with 12 piano students. Tier 1 (immediate): a simple colored indicator on an auxiliary display shows real-time hand position quality using green/yellow/red coding, enabling students to self-correct obvious postural issues without interrupting their playing. Tier 2 (phrase-end): after completing each musical phrase (detected via cadential patterns), the system displays a summary showing detected fingering deviations and their suggested corrections overlaid on a piano roll visualization. Students can tap any flagged note to see the recommended finger number with an optional demonstration video from the training corpus showing expert hand position. Tier 3 (session summary): upon completing a piece, comprehensive feedback includes expressiveness trajectory graphs, comparison with expert reference ranges, and prioritized improvement suggestions.

User study feedback revealed important design considerations. Students initially found simultaneous visual and suggested fingering displays overwhelming; we subsequently implemented a “focus mode” showing only the three most critical corrections per phrase. Intermediate students appreciated seeing alternative valid fingerings rather than single prescriptive corrections, while beginners preferred definitive guidance. These observations suggest that future versions should incorporate adaptive feedback complexity based on detected skill level—a limitation of our current uniform presentation approach.

A notable limitation emerged from our stratified performance analysis: the system occasionally proposes advanced fingering solutions to beginners that, while biomechanically optimal for experienced pianists, exceed the learner's current technical capabilities. For instance, in Hanon exercises, the system sometimes suggested thumb-under passages at tempos requiring finger independence that beginners have not yet developed. This

occurs because our training data distribution, though balanced across skill levels, optimizes for expert-like fingering patterns as ground truth. The model learns to generate what experts would do rather than what pedagogical progression suggests students should learn first.

We attempted to address this through post-hoc filtering based on detected student skill level (inferred from performance accuracy and timing consistency). When skill indicators suggested beginner level, we applied a simplicity preference that penalized fingerings requiring large hand stretches (> 2 octave span) or rapid position shifts (> 2 positions per beat). This heuristic reduced inappropriate advanced suggestions by 34% but introduced a different problem: occasionally rejecting sophisticated fingerings that motivated students were capable of executing.

The fundamental limitation is that our current framework lacks genuine adaptation to individual learners. True personalization would require modeling each student's hand anatomy (finger length ratios, span capability), current technical proficiency (which specific techniques are mastered), and learning trajectory (what they are ready to learn next). Incorporating such student modeling represents a promising direction for future work, potentially drawing on intelligent tutoring system research that maintains learner models updated through interaction history.

Data dependency constitutes a significant limitation requiring candid acknowledgment. Our dataset was collected under controlled laboratory conditions: consistent overhead lighting (5000 K color temperature, 800 lx), fixed camera positions (top-down at 1.2 m height, 45° side angle at 0.8 m), and a single piano model (Yamaha C3 grand). When we evaluated on 150 performances from the MAESTRO dataset—recorded with different pianos, variable lighting, and diverse camera setups—frame accuracy dropped from 89.7% to 82.3%, a 7.4% point degradation.

To systematically characterize this generalization gap, we conducted ablation experiments manipulating individual factors. Changing piano model (acoustic upright vs. our training grand) reduced accuracy by 3.1 points, attributable to different key dimensions affecting hand-keyboard spatial relationships. Varying lighting conditions (dimmer ambient, directional stage lighting) caused 2.8-point degradation due to increased keypoint detection failures. Camera angle variations ($\pm 15^\circ$ from training configuration) produced 4.2-point drops, the largest single factor, because our normalized keypoint representation assumes consistent viewing geometry.

These findings suggest that practical deployment would benefit from domain adaptation techniques. Preliminary experiments with simple input augmentation (random brightness/contrast adjustment, synthetic viewpoint perturbation) during training recovered approximately half the generalization gap without requiring target-domain labeled data. More sophisticated approaches—such as adversarial domain adaptation or self-supervised pretraining on unlabeled piano videos—remain unexplored and represent important future directions.

Computational efficiency considerations extend beyond inference latency to deployment feasibility on resource-constrained devices. Our full model contains 47.2 million parameters occupying 189 MB storage when quantized to 8-bit precision. While acceptable for cloud-based or desktop deployment, this footprint challenges mobile applications where students commonly practice using tablets.

We explored three compression strategies in preliminary experiments. First, knowledge distillation to a smaller student network (12 M parameters) retained 94% of frame accuracy while reducing model size by 75%. Second, structured pruning removing 50% of channels from convolutional layers maintained 91% accuracy with 60% size reduction. Third, INT8 quantization alone (our current deployment configuration) incurred less than 1% accuracy loss while halving memory requirements compared to FP32. Combining distillation with quantization yielded a 15 MB model achieving 85.2% frame accuracy—a reasonable accuracy-efficiency tradeoff for mobile deployment. Recent work on efficient architecture design for visual feature extraction offers complementary strategies that we have not yet explored. Techniques such as pooling-based global guidance and automated architecture search have demonstrated that carefully designed information pathways can achieve strong performance with substantially fewer parameters than conventional backbones—principles that could be adapted to compress our audio-visual encoders specifically for the constrained piano performance domain.

However, the bottleneck extends beyond model size to computational requirements during inference. Real-time hand keypoint detection currently relies on MediaPipe, which itself requires substantial GPU resources for 60fps processing. Edge deployment would require either accepting reduced visual frame rates (degrading accuracy during fast passages) or developing lightweight pose estimation alternatives specifically optimized for the constrained piano performance domain (fixed hand location, limited pose variation compared to general hand tracking).

Generative adversarial networks brought distinct advantages to this multimodal sequence modeling problem. The adversarial training regime encouraged the generator to capture distributional properties of expert performances rather than merely fitting point estimates. This proved valuable for expressiveness modeling where multiple valid interpretations exist—the GAN learned to produce diverse yet plausible suggestions instead of collapsing to a single average solution. However, GANs also introduced instabilities during training. The minimax optimization occasionally stalled when discriminator accuracy saturated prematurely, requiring careful learning rate scheduling and gradient penalty tuning. Mode collapse, though largely mitigated through our multi-objective loss formulation, still occasionally emerged during early training epochs before the expressiveness constraint stabilized the dynamics.

Our approach differs fundamentally from prior work by treating fingering and expressiveness as coupled rather than independent problems. Traditional methods addressed each separately, missing the insight that biomechanically efficient fingerings facilitate expressive freedom. The performance gains stem directly from this holistic perspective, embodied architecturally through the shared multimodal representation that informs both task-specific heads. Additionally, the GAN framework's capacity to model complex conditional distributions

proved superior to classification approaches that reduce fingering choice to a simple labeling problem, ignoring the sequential dependencies and contextual factors that expert performers consider intuitively.

Conclusion

This research investigated the integration of audio-visual features through generative adversarial networks to address two fundamental challenges in computer-assisted piano pedagogy: automated fingering correction and performance expressiveness modeling. We developed a comprehensive framework that treats these traditionally separate problems as coupled tasks benefiting from shared multimodal representations.

Our principal findings demonstrate that cross-modal fusion substantially enhances both fingering accuracy and expressiveness prediction compared to single-modality approaches. The attention-based integration mechanism achieved 89.7% frame-level fingering accuracy, representing meaningful improvement over audio-only and vision-only baselines. Expressiveness modeling attained correlation coefficients exceeding 0.85 for dynamic features and maintained reasonable predictive accuracy even for subtle timbral characteristics. These results validate the hypothesis that acoustic and visual information provide complementary constraints that resolve ambiguities inherent when either modality operates independently.

The methodological innovations introduced in this work span three interconnected dimensions. First, we designed a hierarchical attention architecture that learns adaptive weighting strategies for multimodal fusion rather than relying on fixed combination rules. Second, our conditional GAN formulation treats fingering correction as distribution modeling rather than simple classification, enabling generation of diverse yet plausible suggestions that respect both biomechanical constraints and musical context. Third, the multi-objective loss function balances reconstruction fidelity, adversarial realism, and domain-specific expressiveness criteria through adaptive weight scheduling that prevents any single objective from dominating the optimization landscape.

These contributions hold both theoretical and practical significance for advancing intelligent music education systems. Theoretically, the work demonstrates how deep generative models can capture tacit knowledge about skilled performance that resists explicit rule formulation. Practically, the near real-time inference capability opens possibilities for interactive practice assistance that provides immediate corrective feedback, partially addressing the scarcity of expert instruction.

We must acknowledge several limitations that temper these achievements. The model's dependence on substantial annotated data restricts applicability to new instruments or performance styles without retraining. Generalization across different recording conditions proved less robust than desired, with noticeable performance degradation on out-of-distribution samples. Computational requirements, while manageable for cloud deployment, still challenge edge computing scenarios demanding ultra-low latency responses below 50 milliseconds.

Future research should pursue multiple complementary directions. Constructing larger-scale datasets spanning diverse instruments, performance traditions, and skill levels would improve generalization and reduce annotation bias. Developing lightweight model architectures through neural architecture search and knowledge distillation could enable deployment on resource-constrained mobile devices.

Beyond architectural improvements, emerging generative paradigms offer promising alternatives to our GAN-based approach. Diffusion models have demonstrated superior sample quality and training stability compared to GANs in image and audio generation tasks. Recent work applying diffusion models to multimodal learning—including diffusion-based recovery of missing modality information and hierarchical conditional generation—suggests potential benefits for piano performance modeling. A diffusion-based reformulation of our approach could address two persistent limitations: the mode collapse tendency that occasionally produces repetitive fingering suggestions, and the training instability requiring meticulous hyperparameter tuning. Such a formulation would treat fingering sequence generation as iterative denoising conditioned on audio-visual observations, potentially yielding more diverse and higher-quality outputs. Furthermore, advances in efficient and lightweight attention mechanisms designed for real-time inference suggest pathways toward reducing our current 180ms inference latency, which remains a bottleneck for truly instantaneous note-level feedback. We consider these reformulations a high-priority direction for subsequent investigation. Extending the framework beyond piano to violin, guitar, and other instruments would test whether our fusion principles transfer across different sensorimotor skill domains. Finally, incorporating pedagogical domain knowledge through structured constraints or human-in-the-loop refinement might bridge the gap between technical correctness and teaching appropriateness that currently limits practical adoption.

Data availability

The complete multimodal dataset comprising 3,847 performances (215 h of synchronized audio-visual recordings) is available through a structured access protocol designed to balance reproducibility with privacy protection. Supplementary Material provides all resources necessary for independent replication and is organized as follows: (a) complete feature extraction source code with inline comments, specifying software dependencies (Python 3.9, PyTorch 2.0.1, MediaPipe 0.10.3) and random seeds (seed=42 for all experiments), enabling researchers to replicate our entire preprocessing and training pipeline from raw recordings to final evaluation; (b) configuration files in JSON format listing all hyperparameters for each model component, including learning rate schedules, loss weights (λ_{recon} , λ_{adv} , λ_{expr}), augmentation parameters, and discriminator update frequency rules; (c) evaluation scripts that reproduce all quantitative results reported in Tables 5 and 6, and 7, with expected output values provided for verification; (d) a summary table documenting the detailed architecture specification of every layer in both generator and discriminator networks, including initialization schemes, normalization choices, and dropout placement; (e) sample input-output pairs from 20

representative test performances, showing raw audio-visual features alongside model predictions and ground-truth annotations, to facilitate qualitative inspection without requiring the full dataset. The full dataset access requires completion of a data use agreement acknowledging participant privacy protections and restrictions on redistribution, in accordance with the ethics approval granted by Nanjing Xiaozhuang University (IRB-NJX-ZU-MUS-2024-018). Researchers may request full dataset access by contacting the corresponding author at lijunyu1891299@163.com with a brief description of intended use and institutional affiliation. We commit to responding to access requests within 10 business days. This tiered access approach follows recommendations for responsible sharing of human performance data while maximizing research utility.

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Author contributions

Junyu Li conceived and designed the study, developed the methodology, implemented the experimental framework, conducted data collection and analysis, and wrote the manuscript. The author takes full responsibility for the integrity and accuracy of the work presented.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

This study was approved by the Research Ethics Committee of Nanjing Xiaozhuang University, School of Music (Reference Number: IRB-NJXZU-MUS-2024-018). All participants provided written informed consent prior to enrollment in the recording sessions. The study was conducted in accordance with the Declaration of Helsinki and relevant national regulations of the People's Republic of China. Participants were informed of their right to withdraw from the study at any time without consequence, and all data were anonymized to protect participant privacy.

Consent for publication

The author has reviewed the manuscript and consents to its publication. No identifiable information regarding participants has been included in this manuscript. All participants provided consent for their anonymized performance data to be used for research purposes and potential publication.

Additional information

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